

1. List the members of these sets.
- a) $\{x \mid x \text{ is a real number such that } x^2 = 1\}$
 - b) $\{x \mid x \text{ is a positive integer less than } 12\}$
 - c) $\{x \mid x \text{ is the square of an integer and } x < 100\}$
 - d) $\{x \mid x \text{ is an integer such that } x^2 = 2\}$

a) $\{1, -1\}$
b) $\{1, 2, 3, \dots, 10, 11\}$
c) $\{4, 9, 16, 25, 36, 49, 64, 81\}$
d) $\{\emptyset\}$

2. Use set builder notation to give a description of each of these sets.
- a) $\{0, 3, 6, 9, 12\}$
 - b) $\{-3, -2, -1, 0, 1, 2, 3\}$
 - c) $\{m, n, o, p\}$

a) $\{x \mid x \bmod 3 = 0 \wedge x \leq 12\}$
b) $\{x \mid x \in \mathbb{Z} \wedge -3 \leq x \leq 3\}$
c) $\{x \mid x = m \vee x = n \vee x = o \vee x = p\}$

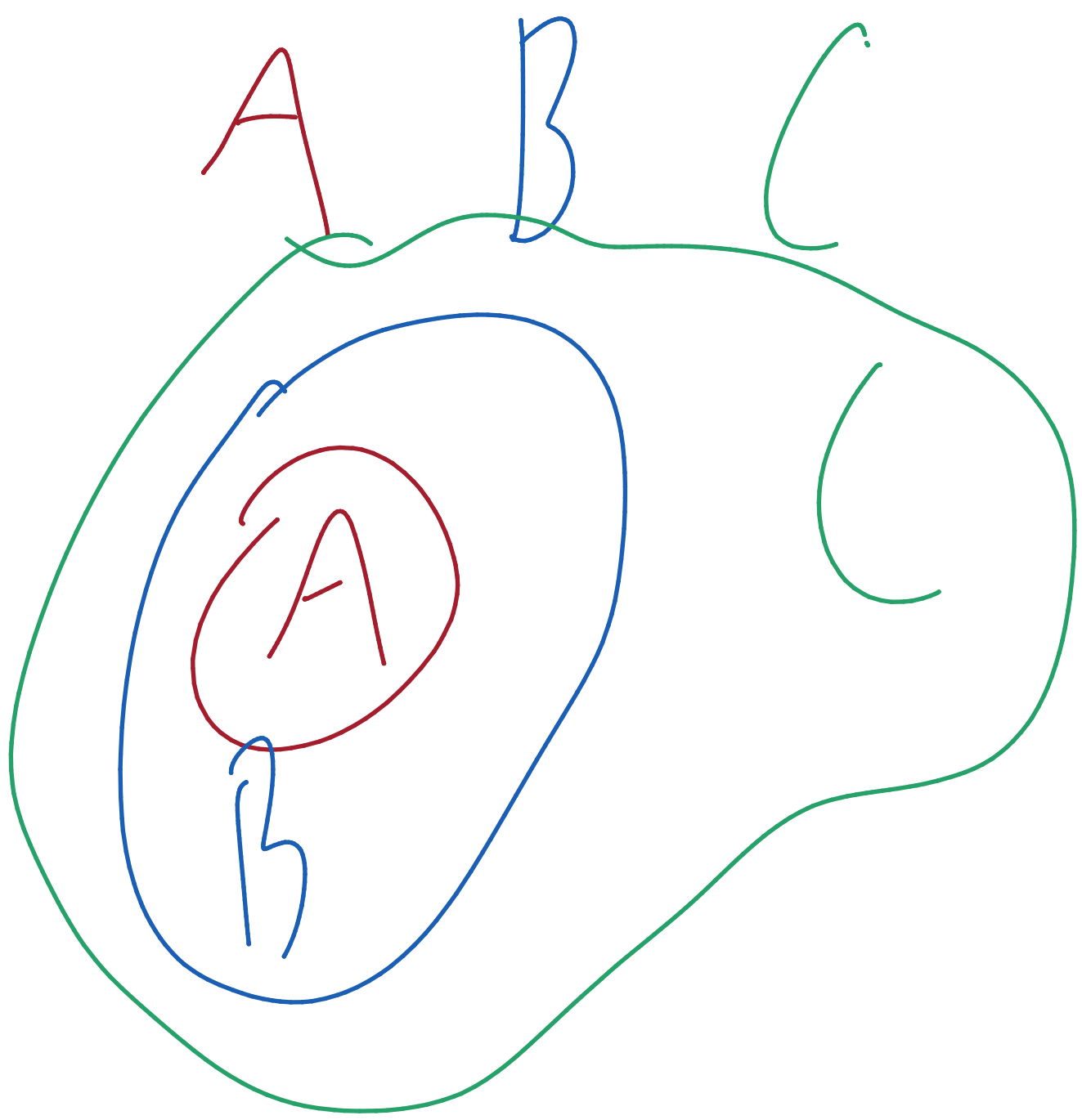
7. Determine whether each of these pairs of sets are equal.

- a) $\{1, 3, 3, 3, 5, 5, 5, 5, 5\}, \{5, 3, 1\}$
 - b) $\{\{1\}\}, \{1, \{1\}\}$ c) $\emptyset, \{\emptyset\}$
- a) equal as comparison is done using distinct members
b) not equal
c) not equal, it's a set of $\emptyset$ as compared to $\emptyset$

12. Determine whether these statements are true or false.
- a) $\emptyset \in \{\emptyset\}$
 - b) $\emptyset \in \{\emptyset, \{\emptyset\}\}$
 - c) $\{\emptyset\} \in \{\emptyset\}$
 - d) $\{\emptyset\} \in \{\{\emptyset\}\}$
 - e) $\{\emptyset\} \subset \{\emptyset, \{\emptyset\}\}$
 - f) $\{\{\emptyset\}\} \subset \{\emptyset, \{\emptyset\}\}$
 - g) $\{\{\emptyset\}\} \subset \{\{\emptyset\}, \{\emptyset\}\}$

- a) T
- b) T
- c) F
- d) T
- e) T
- f) F
- g) F

16. Use a Venn diagram to illustrate the relationship $A \subseteq B$ and $B \subseteq C$.



23. Find the power set of each of these sets, where a and b are distinct elements.

- a) $\{a\}$
- b) $\{a, b\}$
- c) $\{\emptyset, \{\emptyset\}\}$

a) $P(\{a\}) = \{\{a\}, \{\emptyset\}\}$
b) $P(\{a, b\}) = \{\{a, b\}, \{a\}, \{b\}, \{\emptyset\}\}$
c) $P(\{\emptyset, \{\emptyset\}\}) = \{\{\emptyset\}, \{\emptyset, \{\emptyset\}\}, \{\{\emptyset\}\}\}$

2.2

4. Let $A = \{a, b, c, d, e\}$ and $B = \{a, b, c, d, e, f, g, h\}$. Find
- a) $A \cup B$. b) $A \cap B$.
c) $A - B$. d) $B - A$.

a) $A \cup B = \{a, b, c, d, e, f, g, h\}$

b) $A \cap B = \{a, b, c, d, e\}$

c) $A - B = \{\emptyset\}$

d) $B - A = \{f, g, h\}$

In Exercises 5–10 assume that A is a subset of some underlying universal set U .

5. Prove the complementation law in Table 1 by showing that $\overline{\overline{A}} = A$.

$\overline{A}$ is a set such that $A \cap \overline{A} = \emptyset$

and $A \cup \overline{A} = U$

thus $\overline{\overline{A}} \cap A = A$

and $\overline{\overline{A}} = A$

12. Prove the first absorption law from Table 1 by showing that if A and B are sets, then $A \cup (A \cap B) = A$.

$$A \cap B \subseteq A \quad (\text{see \#16a})$$

the union of a set w/ a subset of itself
is itself.

thus $A \cup (A \cap B) = A$

16. Let A and B be sets. Show that

- a) $(A \cap B) \subseteq A$. b) $A \subseteq (A \cup B)$.
c) $A - B \subseteq A$. d) $A \cap (B - A) = \emptyset$.
e) $A \cup (B - A) = A \cup B$.

a) An intersection cannot contain any elements that do not exist in either of its parts. Thus, $A \cap B$ cannot contain any elements not present in A , nor any elements not present in B . As $A \cap B$ contains no elements not found in A , it is a subset of A or equal to A , i.e. $A \subseteq B$.

b) A cannot contain any elements not found in $A \cup B$. This is because a union is the unique sum of its parts, meaning all elements of $A \subseteq (A \cup B)$

c) In subtracting sets, you can get to elements not already present in the first set, in this case set A . Thus $A - B \subseteq A$

d) $B - A$ is all the values of B not contained in A , or more succinctly $B - A = B \cap \overline{A}$. $A \cap \overline{A} = \emptyset$
thus $A \cap (B - A) = A \cap (B \cap \overline{A}) = \emptyset$

e) $B - A = B \cap \overline{A}$ (see above)

$$A \cup (B \cap \overline{A})$$

all the values of B not contained in A
 $B \cap \overline{A}$ are contained in A .

thus $A \cup (B-A) = A \cup B$

21. Show that if A and B are sets, then

- a) $A - B = A \cap \overline{B}$.
- b) $(A \cap B) \cup (A \cap \overline{B}) = A$.

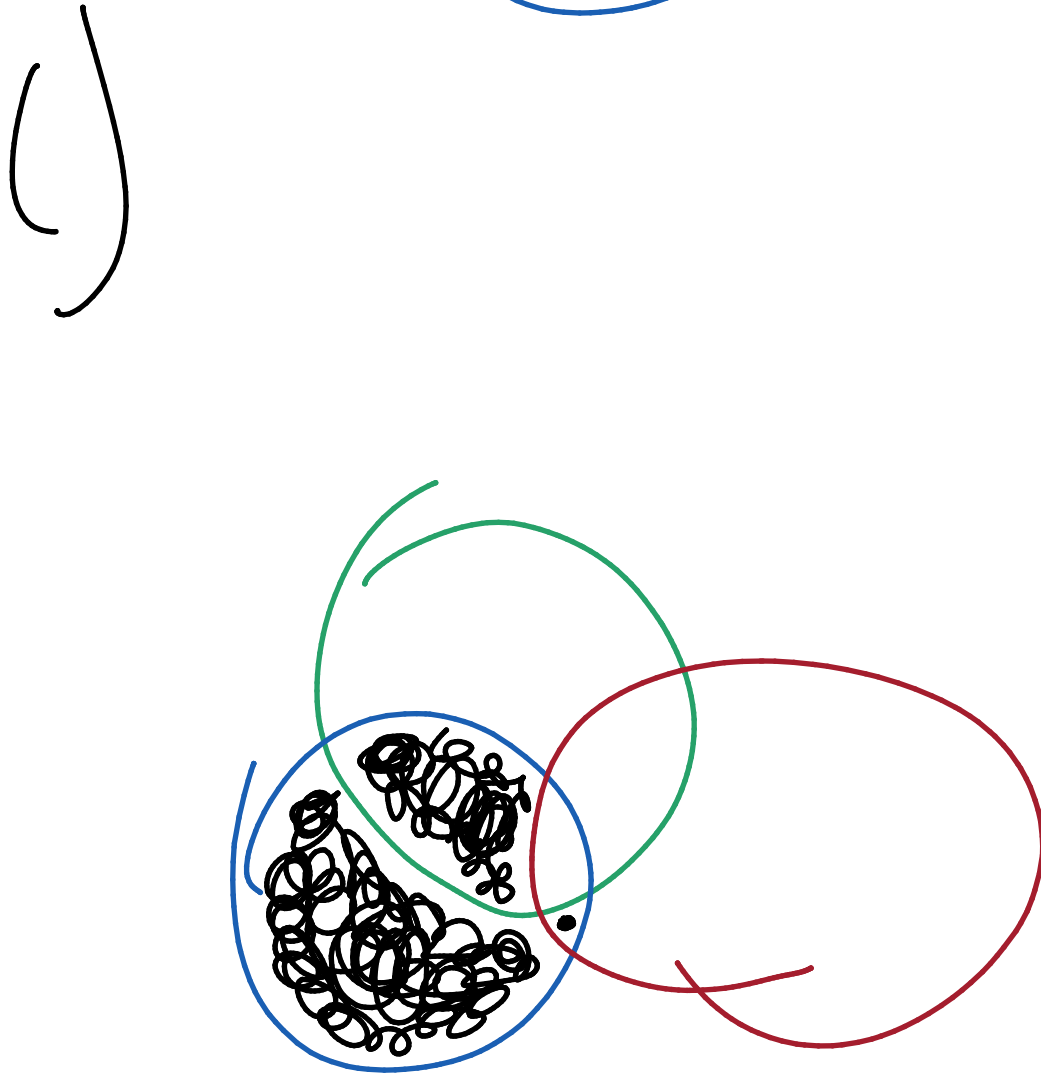
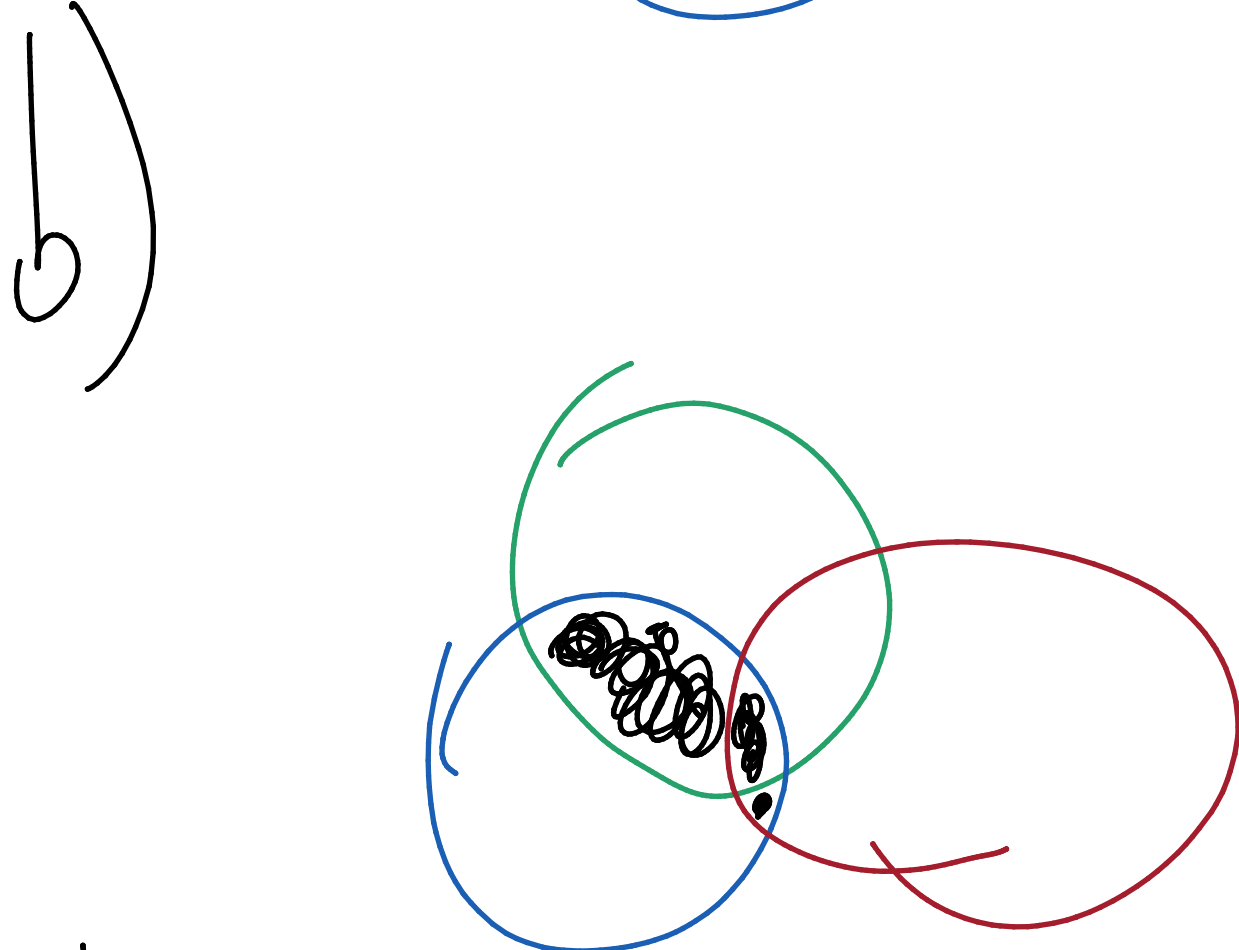
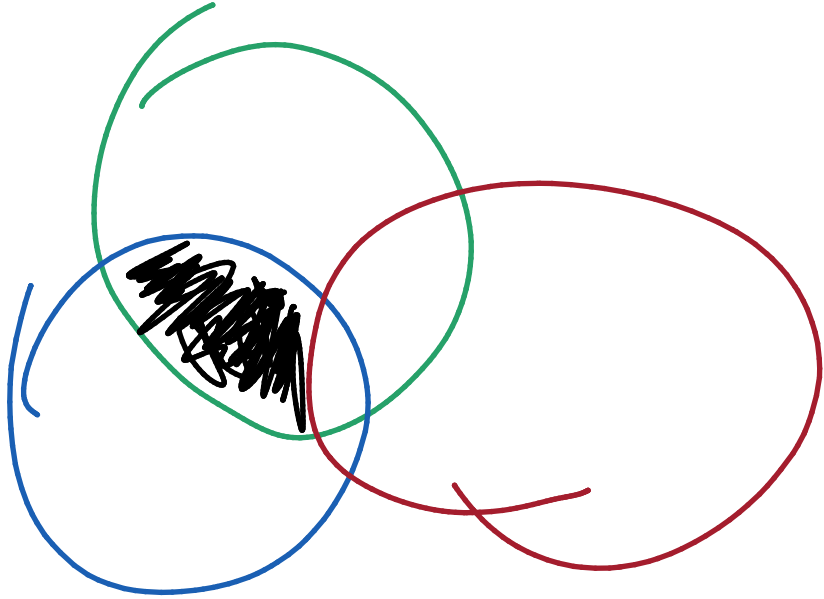
a) $A-B$ represents the values of A not contained in B .
the values not contained in B are represented by $\overline{B}$, thus $A-B=A \cap \overline{B}$

b) $A \cap B$ has all the values in A contained in B .
 $A \cap \overline{B}$ has all the values in A not contained in B .
the union of these are A .
thus, $(A \cap B) \cup (A \cap \overline{B}) = A$

29. Draw the Venn diagrams for each of these combinations of the sets A , B , and C .

- a) $A \cap (B - C)$
- b) $(A \cap B) \cup (A \cap C)$
- c) $(A \cap \overline{B}) \cup (A \cap \overline{C})$

a) A B C



30. Draw the Venn diagrams for each of these combinations of the sets A , B , C , and D .

- a) $(A \cap B) \cup (C \cap D)$
- b) $\overline{A} \cup \overline{B} \cup \overline{C} \cup \overline{D}$
- c) $A - (B \cap C \cap D)$

A B C D

